

- 14** Which of the following statements about the internal energy of a monatomic ideal gas is correct?
- A** It will increase when heat is supplied to the gas.
 - B** It is proportional to the root-mean-square speed of the gas.
 - C** It increases when the temperature of the gas increases.
 - D** It is dependent on the potential energy of the gas.

Ans: C

Since for an ideal gas, internal energy is purely kinetic energy, internal energy is proportional to the temperature of the gas.

- 15** In a mixture of two monatomic ideal gases X and Y , the molecules of Y have thrice the mass